

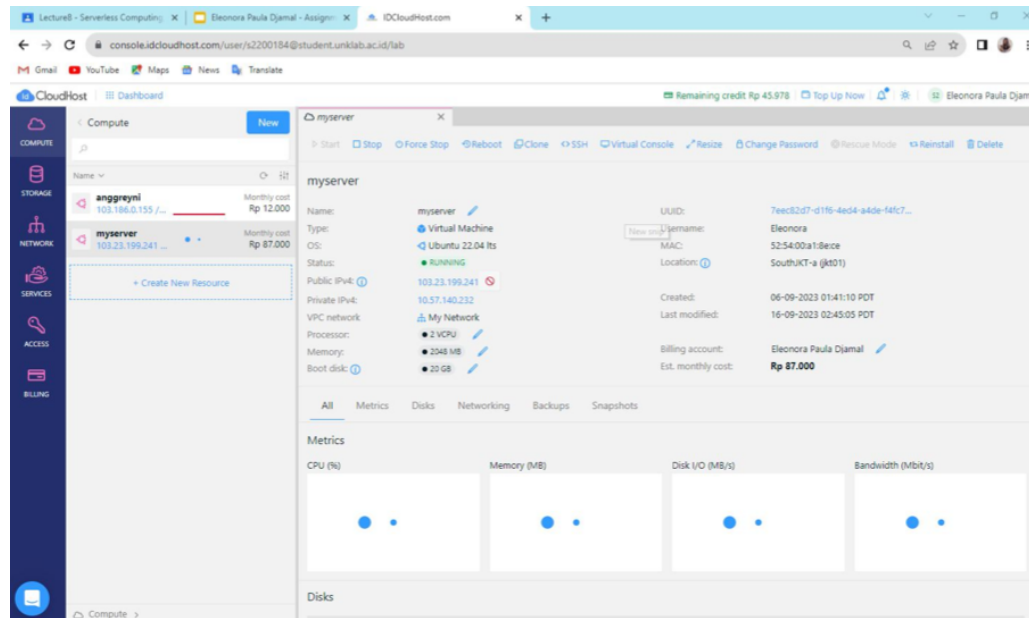
Buat step by step cara development project "Login Form" sampai pada tahapan tranfer ke bagian server side.

Jelaskan dengan detail tahapan untuk membuat domain name di niagahoster.

Jelaskan dengan detail cara untuk menggunakan Let's Encrypt SSL Certificate.

Screenshot dan jelaskan tampilan website yang sudah bisa menggunakan HTTPS url.

NOTE: Untuk development project ini, pastikan bahwa kita sudah membuat Cloud Server terlebih dahulu.



***Tahap pertama yaitu menginstall apache, berikut tahapan menginstall apache2**

[1] Change hak akses superuser atau root:

> sudo su

[2] Perbaharui paket yang tersedia ke versi terbaru ubuntu dengan perintah:

> sudo apt update

> sudo apt upgrade

[3] Install Apache di linux ubuntu 22.04, dengan menjalankan perintah berikut:

> sudo apt install apache2

```
root@myserver:/home/Eleonora
Unpacking file (1:5.41-3ubuntu0.1) over (1:5.41-3) ...
Preparing to unpack .../libmagic1_1:5.41-3ubuntu0.1_amd64.deb ...
Unpacking libmagic1:amd64 (1:5.41-3ubuntu0.1) over (1:5.41-3) ...
Preparing to unpack .../libmagic-mgc_1:5.41-3ubuntu0.1_amd64.deb ...
Unpacking libmagic-mgc (1:5.41-3ubuntu0.1) over (1:5.41-3) ...
Setting up libmagic-mgc (1:5.41-3ubuntu0.1) ...
Setting up gawk (1:5.1.0-ubuntu0.1) ...
Setting up libmagic1:amd64 (1:5.41-3ubuntu0.1) ...
Setting up file (1:5.41-3ubuntu0.1) ...
Setting up open-vm-tools (2:12.1.5-3-ubuntu0.22.04.3) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@myserver:/home/Eleonora# sudo apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.52-1ubuntu4.5).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@myserver:/home/Eleonora#
```

Kita juga bisa cek apakah sudah terinstall atau belum dengan mengakses public ip address milik kita di browser, jika muncul tampilan seperti digambar makanya instalasi apache sukses.



1. Untuk memulai instal MySQL bisa dimulai dengan perintah:

> sudo apt install mysql-server

2. Selanjutnya aktifkan service MySQL di Ubuntu 22.04 dengan perintah:

> sudo systemctl start mysql.service

3. Untuk menambahkan keamanan pada MySQL database, kita bisa menjalankan perintah mysql_secure_installation yang akan menambahkan beberapa pilihan seperti validasi keamanan password, tidak memberikan akses untuk remote login dan membuang sample user.

Disini ada beberapa tahap yang harus kita lakukan, dimulai dari masuk ke MySQL prompt dengan mengetikkan perintah berikut:

> sudo mysql

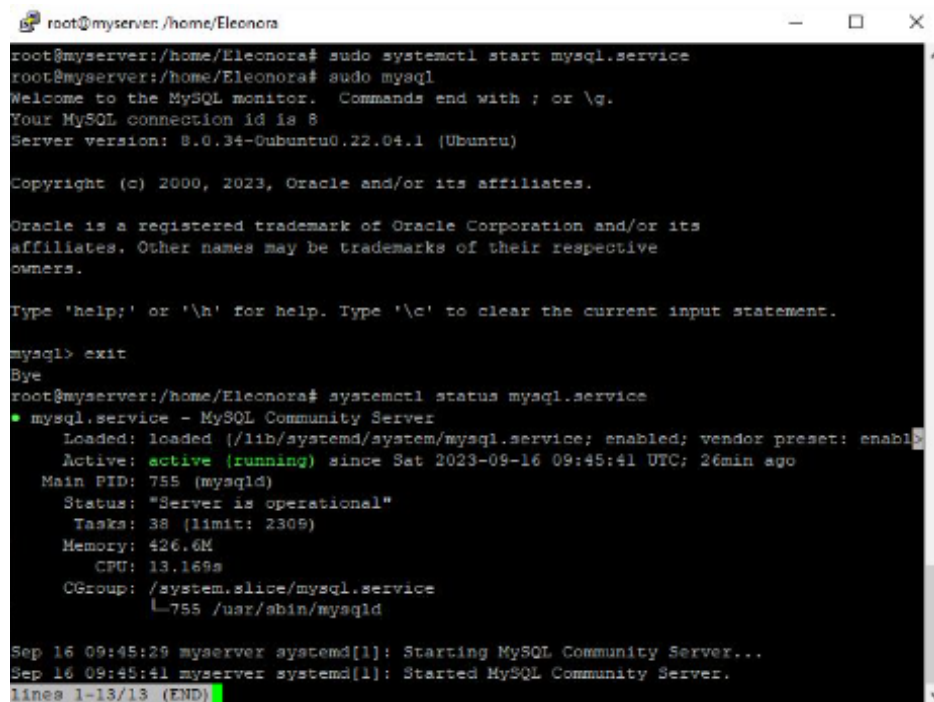
> mysql> ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY 'ganti_password_kita';

Kemudian keluar dari MySQL prompt

mysql> exit

[4] Setelah metode autentikasi untuk user root kita ubah, kemudian kita jalankan perintah berikut:

sudo mysql_secure_installation

A terminal window titled 'root@myserver: /home/Eleonora' showing the execution of several commands. The first command is 'sudo systemctl start mysql.service'. The second is 'sudo mysql', which enters the MySQL prompt. Inside the prompt, the command 'ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY 'ganti_password_kita';' is entered. Then 'exit' is typed to leave the prompt. Finally, 'systemctl status mysql.service' is run, displaying the status of the MySQL service as 'active (running)'. The terminal output includes copyright notices for Oracle and MySQL, and system logs at the bottom showing the service starting at 09:45:29 and 09:45:41.

```
root@myserver: /home/Eleonora
root@myserver:/home/Eleonora# sudo systemctl start mysql.service
root@myserver:/home/Eleonora# sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.34-0ubuntu0.22.04.1 (Ubuntu)

Copyright (c) 2000, 2023, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> exit
Bye
root@myserver:/home/Eleonora# systemctl status mysql.service
● mysql.service - MySQL Community Server
   Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor preset: enable
   Active: active (running) since Sat 2023-09-16 09:45:41 UTC; 26min ago
   Main PID: 755 (mysqld)
   Status: "Server is operational"
   Tasks: 38 (limit: 2309)
   Memory: 426.6M
   CPU: 13.169s
   CGroup: /system.slice/mysql.service
           └─755 /usr/sbin/mysqld

Sep 16 09:45:29 myservers systemd[1]: Starting MySQL Community Server...
Sep 16 09:45:41 myservers systemd[1]: Started MySQL Community Server.
lines 1-13/13 (END)
```

NOTE: ADA BEBERAPA PERTANYAAN YANG AKAN DI JAWAB.

[5] Setelah proses mysql_secure_installation selesai, selanjutnya kita bisa mengembalikan metoda autentikasi user root ke semula yaitu auth_socket dengan beberapa perintah seperti berikut:

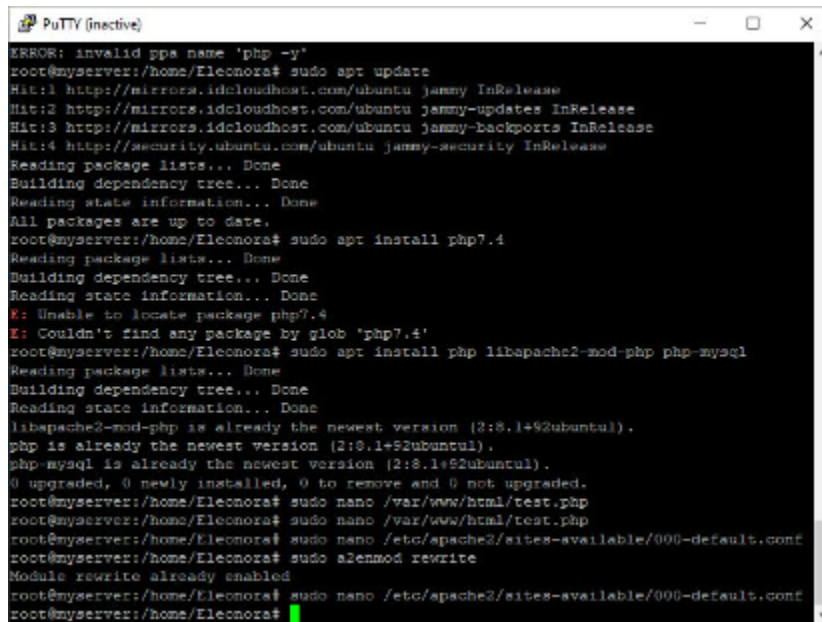
> mysql -u root -p

mysql> ALTER USER 'root'@'localhost' IDENTIFIED WITH auth_socket;

[6] Selanjutnya, kita bisa melihat status MySQL service dengan perintah: systemctl status mysql.service

[1] Menginstall PHP sekaligus modul pendukung lainnya:

> sudo apt install php libapache2-mod-php php-mysql



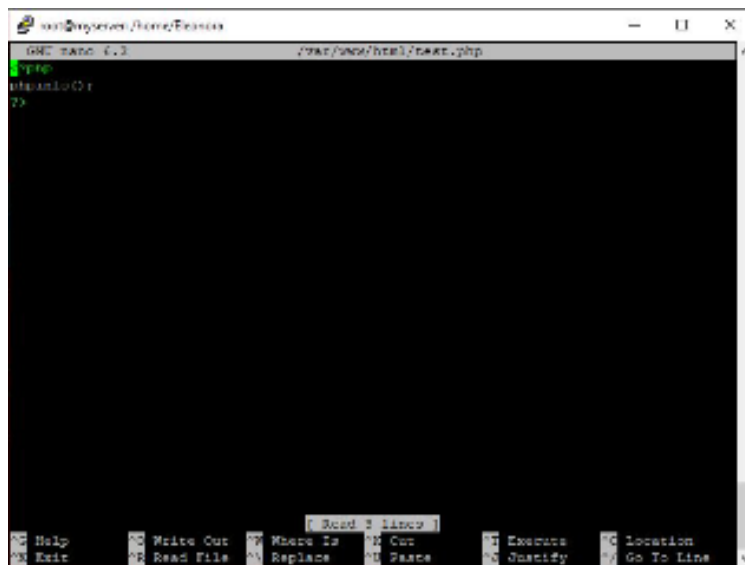
```
ERROR: invalid gpg name 'php -y'
root@myserver:/home/Elecnora# sudo apt update
Hit:1 http://mirrors.idcloudhost.com/ubuntu jammy InRelease
Hit:2 http://mirrors.idcloudhost.com/ubuntu jammy-updates InRelease
Hit:3 http://mirrors.idcloudhost.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
root@myserver:/home/Elecnora# sudo apt install php7.4
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package php7.4
E: Couldn't find any package by glob 'php7.4'
root@myserver:/home/Elecnora# sudo apt install php libapache2-mod-php php-mysql
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
libapache2-mod-php is already the newest version (2:8.1+52ubuntu1).
php is already the newest version (2:8.1+52ubuntu1).
php-mysql is already the newest version (2:8.1+52ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@myserver:/home/Elecnora# sudo nano /var/www/html/test.php
root@myserver:/home/Elecnora# sudo nano /var/www/html/test.php
root@myserver:/home/Elecnora# sudo nano /etc/apache2/sites-available/000-default.conf
root@myserver:/home/Elecnora# sudo a2enmod rewrite
Module rewrite already enabled
root@myserver:/home/Elecnora# sudo nano /etc/apache2/sites-available/000-default.conf
root@myserver:/home/Elecnora#
```

[2] Buat “test.php” file guna memastikan PHP sudah terinstall di server:

> sudo nano /var/www/html/test.php

Dan ketikan:

```
<?php
phpinfo();
?>
```



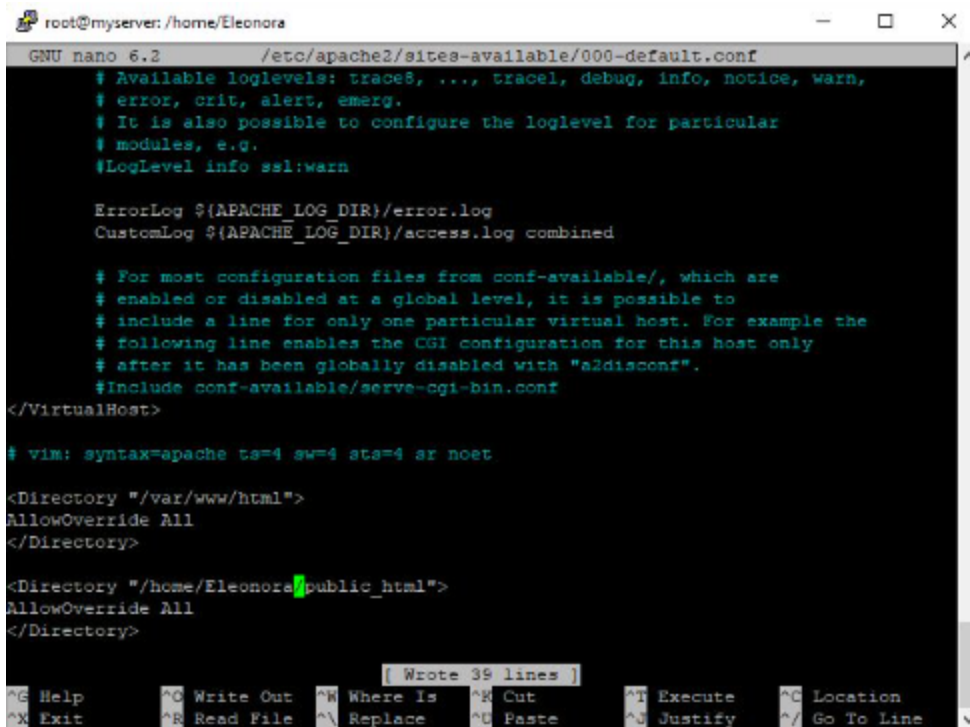
```
GNU nano 4.3 /var/www/html/test.php
?php
phpinfo();
?>
```

[3] Akhiri dengan klik tombol Ctrl + X dan type Y, lalu Enter secara berurutan.

[4] Silakan kembali ke browser dan ketikan “https://public-ip-address/test.php” (contoh: h103.23.198.38).

Browser akan memunculkan tampilan PHP version yang berarti kita telah sukses menginstall php di Ubuntu

</Directory>



```
root@myserver: /home/Eleonora
GNU nano 6.2 /etc/apache2/sites-available/000-default.conf
# Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
# error, crit, alert, emerg.
# It is also possible to configure the loglevel for particular
# modules, e.g.
#LogLevel info ssl:warn

ErrorLog ${APACHE_LOG_DIR}/error.log
CustomLog ${APACHE_LOG_DIR}/access.log combined

# For most configuration files from conf-available/, which are
# enabled or disabled at a global level, it is possible to
# include a line for only one particular virtual host. For example the
# following line enables the CGI configuration for this host only
# after it has been globally disabled with "a2disconf".
#Include conf-available/serve-cgi-bin.conf
</VirtualHost>

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet

<Directory "/var/www/html">
AllowOverride All
</Directory>

<Directory "/home/Eleonora/public_html">
AllowOverride All
</Directory>

Wrote 39 lines
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^N Replace   ^P Paste     ^J Justify  ^_ Go To Line
```

[5] Simpan dan tutup file konfigurasi dengan menekan Ctrl+X, lalu ketik Y untuk menyimpan perubahan dan Enter untuk menutup editor.

[6] Restart apache web server:

```
> sudo systemctl restart apache2
```

Menginstall phpMyAdmin di Ubuntu 22.04

Berikut tahapan untuk menginstall phpMyAdmin di Ubuntu

```
> sudo su
```

```
> sudo apt install unzip
```

```
> wget https://files.phpmyadmin.net/phpMyAdmin/5.2.0/phpMyAdmin-5.2.0-all-languages.zip
```

```
> unzip phpMyAdmin-5.2.0-all-languages.zip
```

```
> sudo mv phpMyAdmin-5.2.0-all-languages /usr/share/phpmyadmin
```

Setelah sukses menginstall, langkah selanjutnya konfigurasi file phpMyAdmin.

Tulis command berikut untuk konfigurasi file.

```
> sudo vim /etc/apache2/conf-available/phpmyadmin.conf
```

Setelah masuk ke dalam file, isi teks konfigurasi berikut:

Jangan lupa untuk restart apache

```
> sudo a2enconf phpmyadmin
```

```
> sudo systemctl restart apache2
```

```
> sudo systemctl reload apache2
```

```
> sudo systemctl status apache2.service
```

Membuat MySQL Database dan User

Pada tahap awal silahkan masuk ke MySQL server:

```
> sudo mysql  
> mysql -u root -p
```

Selanjutnya buat user dan juga database:

Note*: Ubah user "testadmin" dengan nama user sendiri.

Untuk password pada bagian IDENTIFIED BY 'NewPa\$\$w0rd'; wajib membuat password masing-masing.

```
mysql> CREATE DATABASE testadmin;
```

```
mysql> CREATE USER 'testadmin'@'localhost' IDENTIFIED BY 'NewPa$$w0rd';
```

```
mysql> GRANT ALL ON testadmin.* TO 'testadmin'@'localhost';
```

```
mysql> GRANT ALL PRIVILEGES ON *.* TO 'testadmin'@'localhost';
```

```
mysql> FLUSH PRIVILEGES;
```

Setelah itu akses phpmyadmin menggunakan IP address seperti berikut:

```
> http://your-server-ip-domain/phpmyadmin
```

(karena pada bagian ini saya error, saya menggunakan cara manual seperti yg dibawah ini)

NOTE* Ada case dimana kita tidak bisa mengakses halaman web phpmyadmin menggunakan IP address

Untuk mengakali hal tersebut, kita bisa mengisi data pada Database yang telah dibuat sebelumnya.

Langkah awal yaitu masuk ke dalam server MySQL.\

```
mysql> CREATE DATABASE db_unklab;
```

```
mysql> SHOW DATABASES;
```

```
mysql> use db_unklab;
```

SETELAH ITU BUAT TABEL DAN ISI DATA PADA TABEL YANG TELAH DIBUAT.

```
mysql> CREATE TABLE tbl_students (reg_number VARCHAR(255), nim_number VARCHAR(255), email  
VARCHAR(255), fullname VARCHAR(255), password VARCHAR(255), created_at VARCHAR(255), updated_at  
VARCHAR(255));
```

```
mysql> INSERT INTO tbl_students (reg_number, nim_number, email, fullname, password, created_at, updated_at)  
VALUES
```

```

-> ('1001', '1234567890', 'john@example.com', 'John Doe', 'password123', NOW(), NOW()),
-> ('1002', '0987654321', 'jane@example.com', 'Jane Smith', 'password456', NOW(), NOW()),
-> ('1003', '2468013579', 'bob@example.com', 'Bob Johnson', 'password789', NOW(), NOW()),
-> ('1004', '1357902468', 'alice@example.com', 'Alice Brown', 'passwordabc', NOW(), NOW()),
-> ('1005', '3141592653', 'jim@example.com', 'Jim Black', 'passworddef', NOW(), NOW());

```

```

mysql> CREATE TABLE tbl_operator (nip VARCHAR(255), email VARCHAR(255), fullname VARCHAR(255), password
VARCHAR(255), phone_number VARCHAR(255), role VARCHAR(255), created_at VARCHAR(255), updated_at
VARCHAR(255));

```

```

mysql> INSERT INTO tbl_operator (nip, email, fullname, password, phone_number, role, created_at, updated_at)
-> VALUES
-> ('1234567890', 'john@example.com', 'John Doe', 'password123', '081234567890', 'admin', NOW(), NOW()),
-> ('0987654321', 'jane@example.com', 'Jane Smith', 'password456', '087654321098', 'staff', NOW(), NOW()),
-> ('2468013579', 'bob@example.com', 'Bob Johnson', 'password789', '082468013579', 'staff', NOW(), NOW()),
-> ('1357902468', 'alice@example.com', 'Alice Brown', 'passwordabc', '081357902468', 'staff', NOW(), NOW()),
-> ('3141592653', 'jim@example.com', 'Jim Black', 'passworddef', '083141592653', 'admin', NOW(), NOW());

```

Jika sudah, silahkan cek apakah semua data telah terisi.
Sebagai contoh pada gambar:

```
eleonorad Jamal — root@myserver: /home/Eleonora — ssh Eleonora@103.23.199.241 — 129x45

mysql> INSERT INTO tbl_operator (nip, email, fullname, password, phone_number, role, created_at, updated_at)
-> VALUES
-> ('1234567890', 'john@example.com', 'John Doe', 'password123', '081234567890', 'admin', NOW(), NOW()),
-> ('0987654321', 'jane@example.com', 'Jane Smith', 'password456', '087654321098', 'staff', NOW(), NOW()),
-> ('2468013579', 'bob@example.com', 'Bob Johnson', 'password789', '082468013579', 'staff', NOW(), NOW()),
-> ('1357902468', 'alice@example.com', 'Alice Brown', 'passwordabc', '081357902468', 'staff', NOW(), NOW()),
-> ('3141592653', 'jim@example.com', 'Jim Black', 'passworddef', '083141592653', 'admin', NOW(), NOW());
Query OK, 5 rows affected (0.22 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SHOW TABLES;
+-----+
| Tables_in_db_unklab |
+-----+
| tbl_operator         |
| tbl_students         |
+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM tbl_operator;
+-----+-----+-----+-----+-----+-----+-----+-----+
| nip      | email      | fullname | password | phone_number | role  | created_at | updated_at |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1234567890 | john@example.com | John Doe | password123 | 081234567890 | admin | 2023-11-21 11:09:35 | 2023-11-21 11:09:35 |
| 0987654321 | jane@example.com | Jane Smith | password456 | 087654321098 | staff | 2023-11-21 11:09:35 | 2023-11-21 11:09:35 |
| 2468013579 | bob@example.com | Bob Johnson | password789 | 082468013579 | staff | 2023-11-21 11:09:35 | 2023-11-21 11:09:35 |
| 1357902468 | alice@example.com | Alice Brown | passwordabc | 081357902468 | staff | 2023-11-21 11:09:35 | 2023-11-21 11:09:35 |
| 3141592653 | jim@example.com | Jim Black | passworddef | 083141592653 | admin | 2023-11-21 11:09:35 | 2023-11-21 11:09:35 |
+-----+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

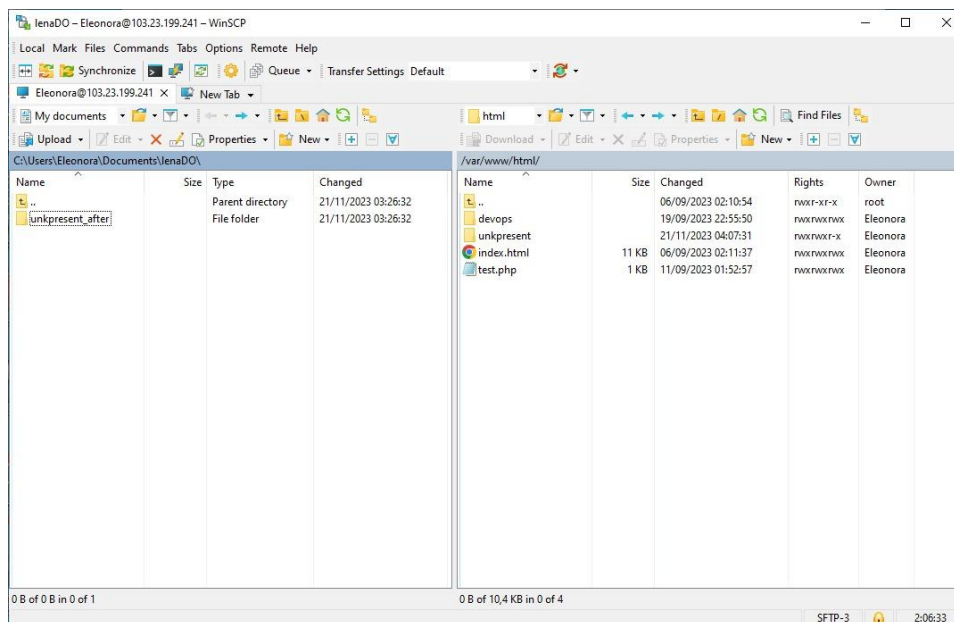
mysql> SELECT * FROM tbl_students;
+-----+-----+-----+-----+-----+-----+-----+
| reg_number | nim_number | email      | fullname | password | created_at | updated_at |
+-----+-----+-----+-----+-----+-----+-----+
| 1001       | 1234567890 | john@example.com | John Doe | password123 | 2023-11-21 11:09:03 | 2023-11-21 11:09:03 |
| 1002       | 0987654321 | jane@example.com | Jane Smith | password456 | 2023-11-21 11:09:03 | 2023-11-21 11:09:03 |
| 1003       | 2468013579 | bob@example.com | Bob Johnson | password789 | 2023-11-21 11:09:03 | 2023-11-21 11:09:03 |
| 1004       | 1357902468 | alice@example.com | Alice Brown | passwordabc | 2023-11-21 11:09:03 | 2023-11-21 11:09:03 |
| 1005       | 3141592653 | jim@example.com | Jim Black | passworddef | 2023-11-21 11:09:03 | 2023-11-21 11:09:03 |
+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

Selanjutnya memasukan project “login form” ke server side.

Cara untuk transfer project login form bisa kita permudah dengan menggunakan tools WinSCP. Masukan folder **unkpresent** yang telah disiapkan ke dalam directory `/var/www/html/`.

Setelah folder dimasukkan, jangan lupa konfigurasi file `/var/www/html/unkpresent/app/config/config.php`.



(atau cara manual untuk transfer file)

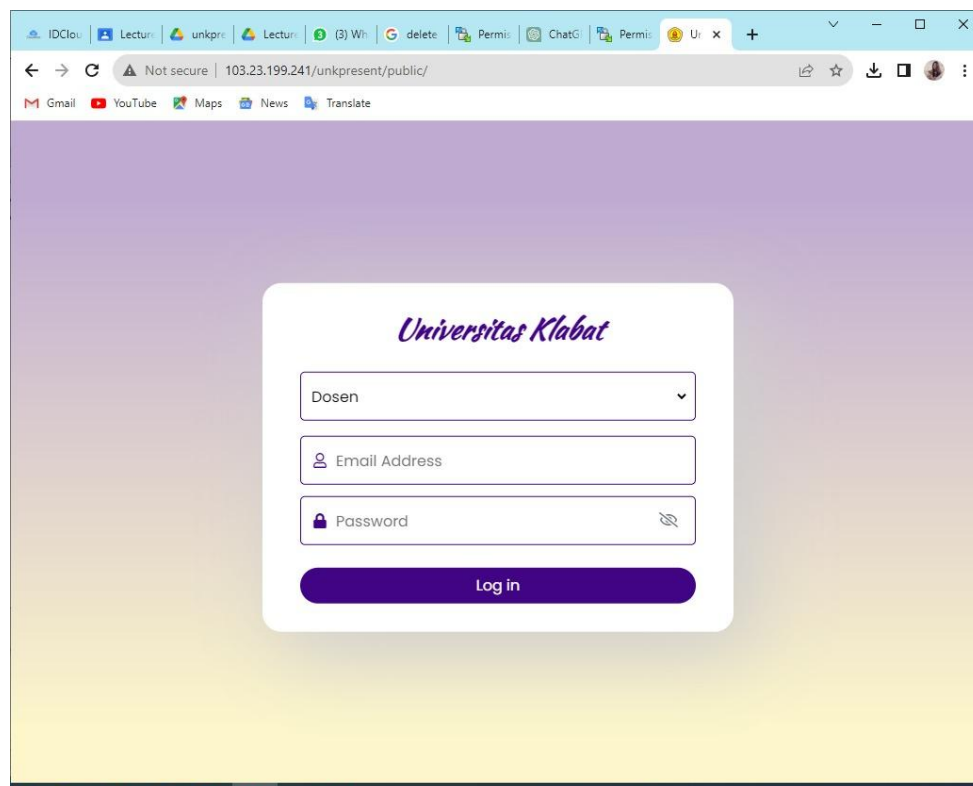
```
Command Prompt
Eleonora@103.23.199.241's password:
scp: /var/www/html/unkpresent: Permission denied

C:\Users\Eleonora>scp -r C:\Users\Eleonora\Documents\lenaDO\unkpresent_after\unkpresent Eleonora@103.23.199.241:\var\www
\html
Eleonora@103.23.199.241's password:
. htaccess 100% 16 0.2KB/s 00:00
config.php 100% 299 4.7KB/s 00:00
home.php 100% 1003 12.5KB/s 00:00
login.php 100% 1385 12.4KB/s 00:00
project.php 100% 312 2.4KB/s 00:00
App.php 100% 2476 22.1KB/s 00:00
Controller.php 100% 462 5.8KB/s 00:00
Database.php 100% 938 14.7KB/s 00:00
init.php 100% 263 4.1KB/s 00:00
Login_model.php 100% 984 15.4KB/s 00:00
default.php 100% 592 9.3KB/s 00:00
page.php 100% 267 4.2KB/s 00:00
index.php 100% 3254 50.8KB/s 00:00
index.php 100% 198 3.1KB/s 00:00
footer.php 100% 466 7.3KB/s 00:00
header.php 100% 1058 16.5KB/s 00:00
. htaccess 100% 154 2.4KB/s 00:00
bootstrap.min.css 100% 152KB 204.5KB/s 00:00
login-style.css 100% 3402 53.3KB/s 00:00
test.png 100% 696KB 400.7KB/s 00:01
index.php 100% 110 1.7KB/s 00:00
bootstrap.bundle.js 100% 203KB 357.8KB/s 00:00
bootstrap.bundle.min.js 100% 77KB 410.2KB/s 00:00
bootstrap.esm.js 100% 136KB 414.4KB/s 00:00
bootstrap.esm.min.js 100% 71KB 324.6KB/s 00:00
bootstrap.js 100% 145KB 422.2KB/s 00:00
bootstrap.min.js 100% 59KB 270.9KB/s 00:00
jquery-3.3.1.slim.min.js 100% 68KB 377.8KB/s 00:00
login-js-handling.js 100% 1091 17.1KB/s 00:00

C:\Users\Eleonora>
```

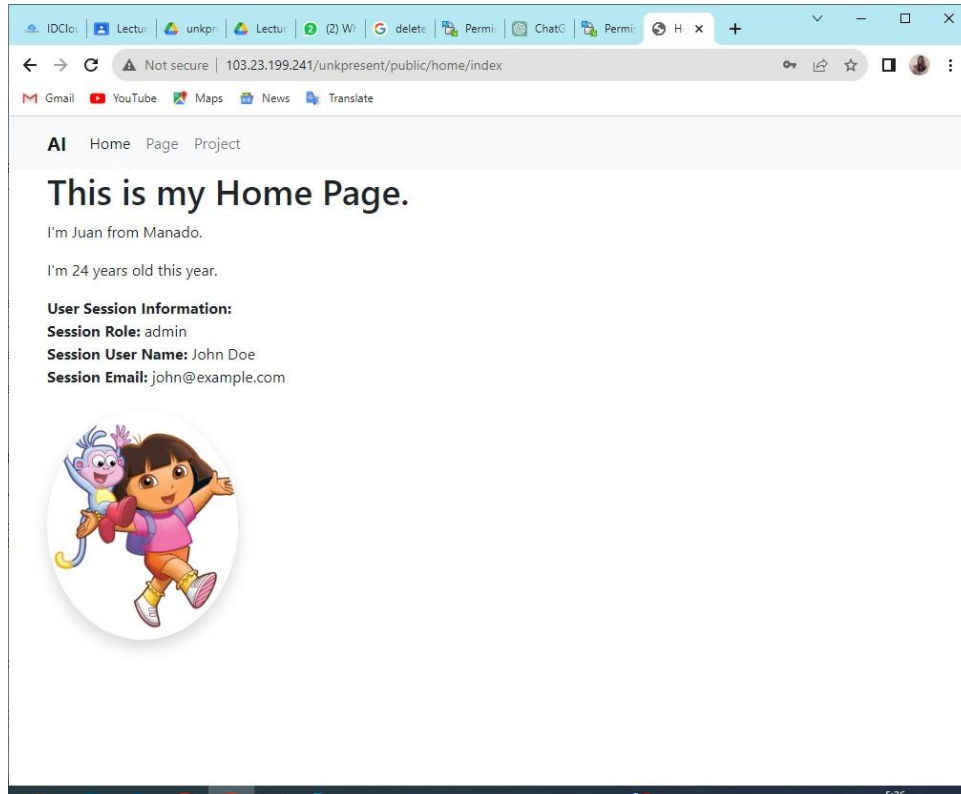
Selanjutnya kita coba akses login form yang telah kita masukkan.

Output yang dihasilkan seperti berikut:



Kita coba login menggunakan salah satu data yang sudah kita isi sebelumnya.

Output jika berhasil login:



2 – Membuat domain name di niagahoster.

Tahap awal membuat domain name:

Akses <https://www.niagahoster.co.id/domain-murah> kemudian sign in.

Masukkan nama domain yang diinginkan lalu cari harga yang paling murah, sebagai contoh cari “eleonorapaula.online” karena .online yang murah.

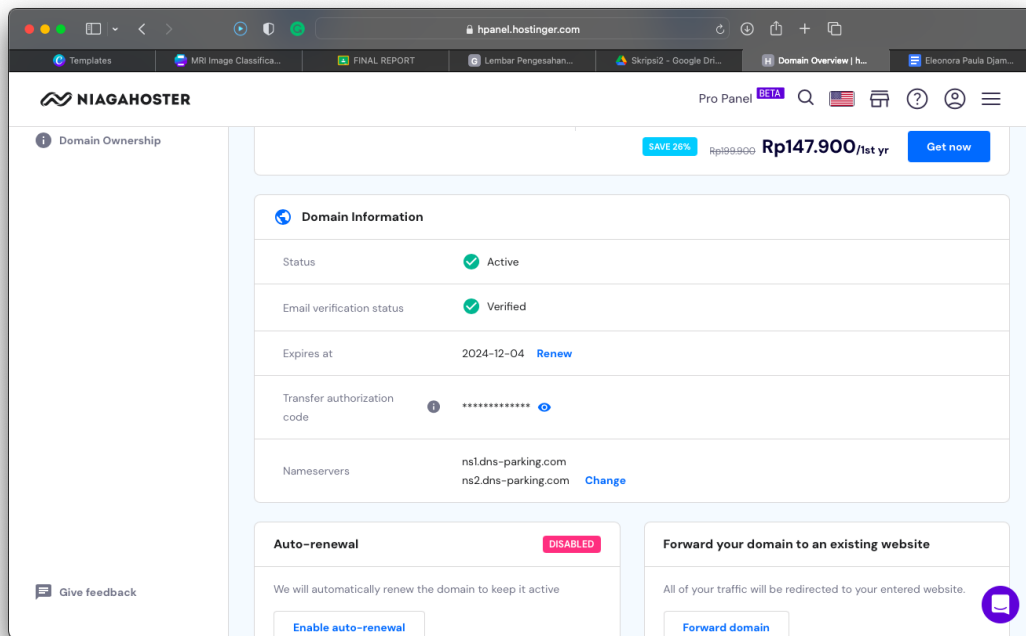
Setelah mendapat domain murah, klik “Pilih” untuk melanjutkan proses pembayaran.

Pilih durasi registrasi domain

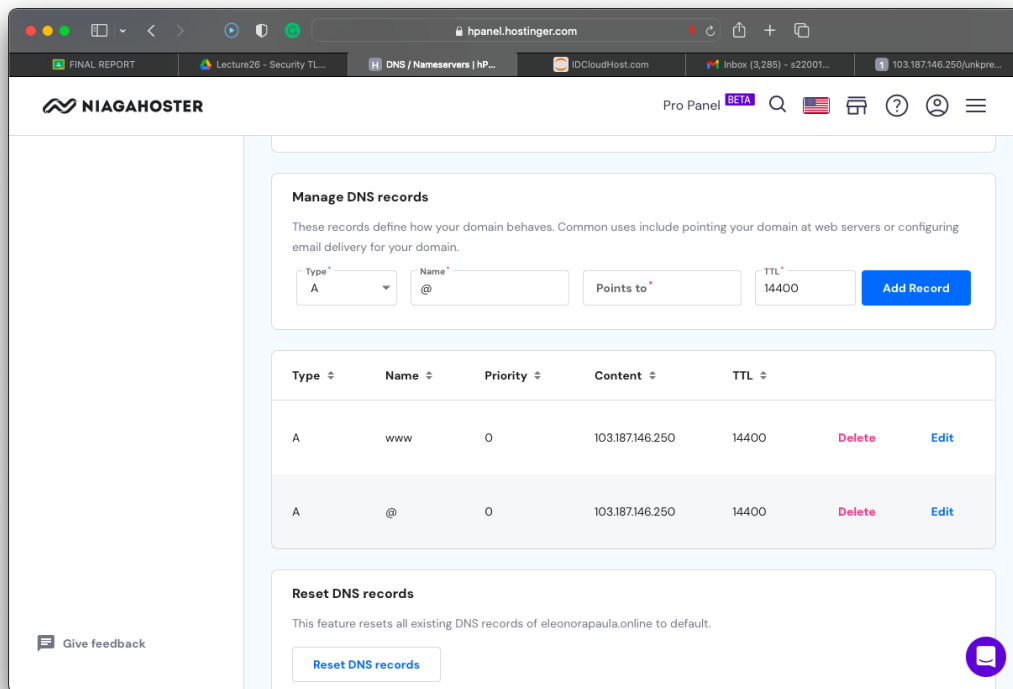
Pilih metode pembayaran

Setelah itu masuk ke dashboard Niagahoster, cek apakah sudah aktif. Jika sudah aktif selanjutnya klik “Manage”

Pastikan sudah verified by Email.



klik Tab “DNS/Nameservers” lalu ganti menggunakan IP address masing-masing.



3 – Menggunakan Let's Encrypt SSL Certificate.

Tahap awal untuk menggunakan Let's Encrypt SSL Certificate:

Gunakan command berikut:

- > sudo apt-get update
- > sudo snap install --classic certbot
- > sudo certbot --help
- > sudo certbot --apache



```
eleonoradjamal — elen@elenserver: ~ — ssh elen@103.187.146.250 — 106x55
Certbot can obtain and install HTTPS/TLS/SSL certificates. By default,
it will attempt to use a webserver both for obtaining and installing the
certificate.
certbot: error: unrecognized arguments: --apache
elen@elenserver:~$ sudo certbot --apache
usage:
  certbot [SUBCOMMAND] [options] [-d DOMAIN] [-d DOMAIN] ...

Certbot can obtain and install HTTPS/TLS/SSL certificates. By default,
it will attempt to use a webserver both for obtaining and installing the
certificate.
certbot: error: unrecognized arguments: --apache
elen@elenserver:~$ sudo certbot --apache
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Enter email address (used for urgent renewal and security notices)
(Enter 'c' to cancel): s2200184@student.unklab.ac.id

-----
Please read the Terms of Service at
https://letsencrypt.org/documents/LE-SA-v1.3-September-21-2022.pdf. You must
agree in order to register with the ACME server. Do you agree?
-----
(Y)es/(N)o: Y

-----
Would you be willing, once your first certificate is successfully issued, to
share your email address with the Electronic Frontier Foundation, a founding
partner of the Let's Encrypt project and the non-profit organization that
develops Certbot? We'd like to send you email about our work encrypting the web,
EFF news, campaigns, and ways to support digital freedom.
-----
(Y)es/(N)o: Y
Account registered.
Please enter the domain name(s) you would like on your certificate (comma and/or
space separated) (Enter 'c' to cancel): eleonorapaula.online
Requesting a certificate for eleonorapaula.online

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/eleonorapaula.online/fullchain.pem
Key is saved at: /etc/letsencrypt/live/eleonorapaula.online/privkey.pem
This certificate expires on 2024-03-02.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

Deploying certificate
Successfully deployed certificate for eleonorapaula.online to /etc/apache2/sites-available/000-default-le-
ssl.conf
Congratulations! You have successfully enabled HTTPS on https://eleonorapaula.online

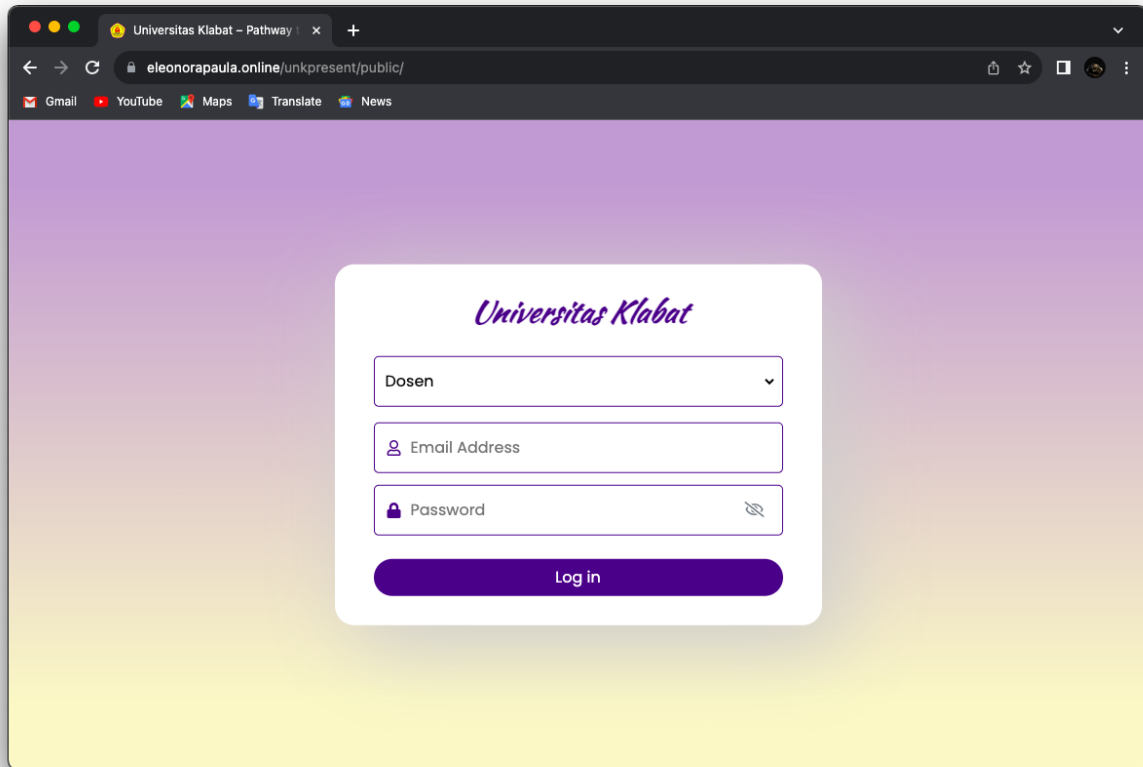
-----
If you like Certbot, please consider supporting our work by:
* Donating to ISRG / Let's Encrypt: https://letsencrypt.org/donate
* Donating to EFF: https://eff.org/donate-le
-----
elen@elenserver:~$
```

4 – Tampilan website yang sudah bisa menggunakan HTTPS url.

Setelah membuat domain dan menggunakan Let's Encrypt SSL Certificate, kita dapat mengakses menggunakan https dan juga domain yang telah kita buat. Contoh pada gambar dibawah:

Ini adalah tampilan website yang sudah bisa menggunakan HTTPS url.
Kita juga dapat mengaksesnya hanya menggunakan www maupun domain.

Sebelum mengakses Login Form, kita harus konfigurasi kembali menggunakan tool WinSCP pada directory dan file
“/var/www/html/unkpresent/app/config/config.php”



Dan ini hasil akhirnya terlihat ada gambar kunci maka ini artinya bahwa web sudah secure